Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. We trained each model for 30 epochs with a batch size of 64.

2 Results

Figure 1 shows the test accuracy by dataset.

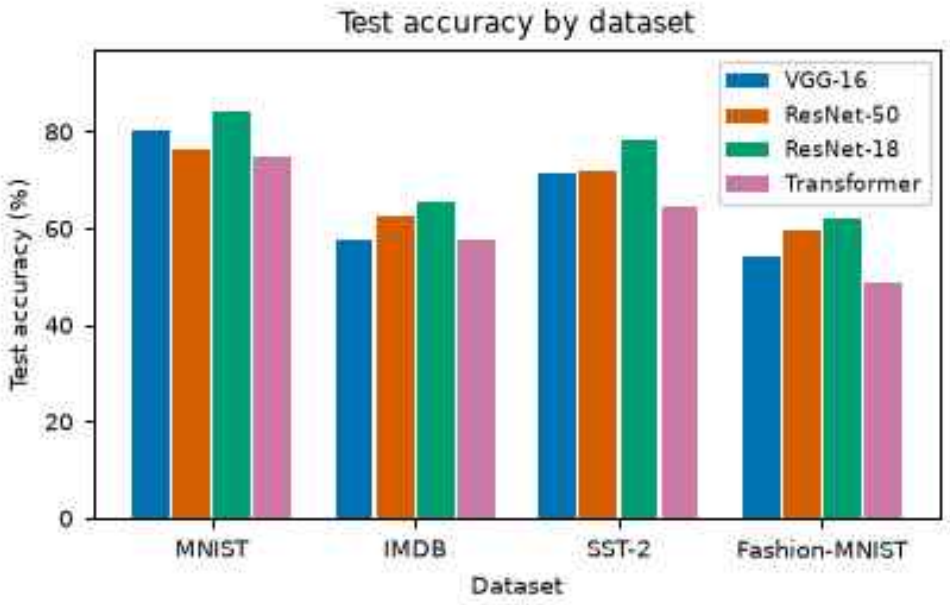


Figure 1: Test accuracy by dataset for the models in this experiment.

On SST-2, ResNet-18 is 15.8 points better than Transformer. As shown in Figure 1, on MNIST, ResNet-18 is 27.2 points better than ResNet-50.

Figure 2 shows the f1 score of EfficientNet by dataset.

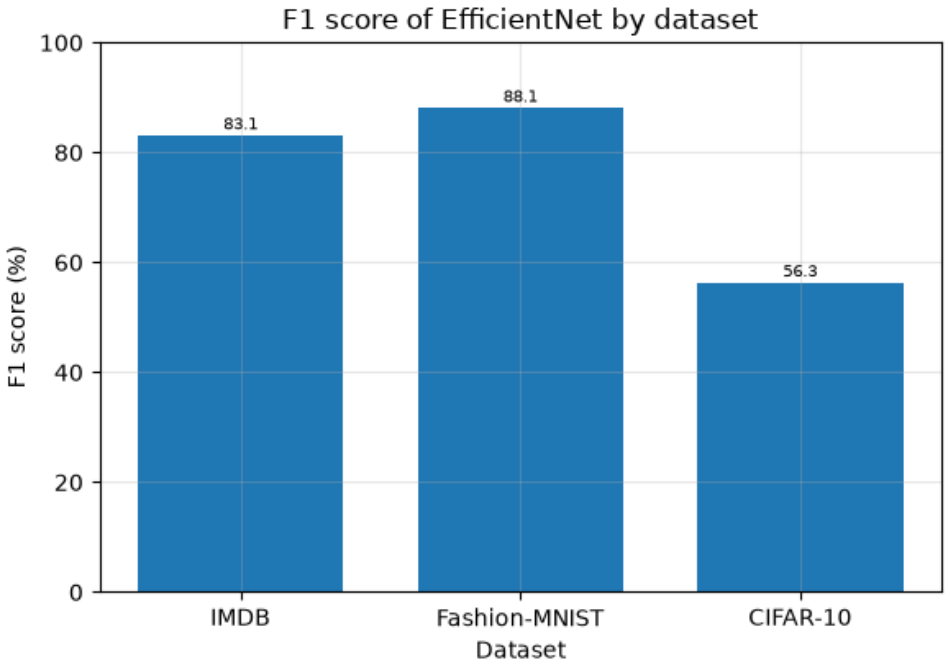


Figure 2: F1 score of EfficientNet by dataset for the models in this experiment.

EfficientNet scores 31.8 points higher on Fashion-MNIST than on CIFAR-10. EfficientNet scores 5.0 points

higher on Fashion-MNIST than on IMDB.

Figure 3 shows the accuracy of Ours by dataset.

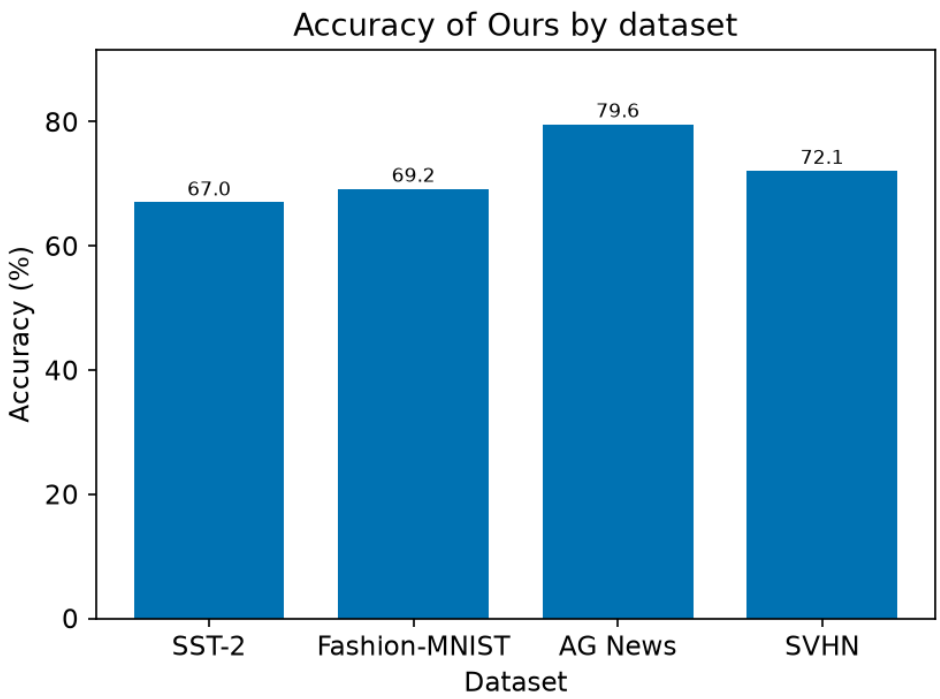


Figure 3: Accuracy of Ours by dataset for the models in this experiment.

Ours scores 4.7 points higher on SVHN than on Fashion-MNIST. On SVHN, Ours obtains a accuracy of 72.1%. Ours scores 10 points higher on AG News than on Fashion-MNIST.

Figure 4 shows the accuracy by dataset.

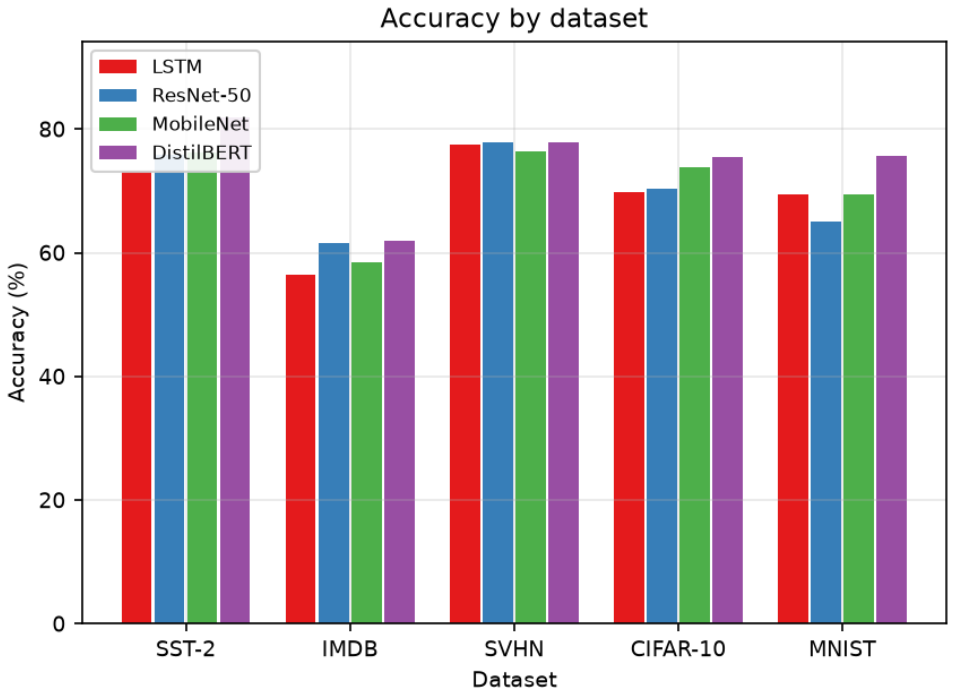


Figure 4: Accuracy by dataset for the models in this experiment.

LSTM reaches 73.9% accuracy on CIFAR-10 (Figure 4). DistilBERT outperforms MobileNet by 1.4 percentage points on SVHN.

3 Discussion

SVM reaches 73.9% accuracy on CIFAR-10. We note that the training was stable for all models.